

Advanced Optimization

Chapter 6 — Quasi-Newton Methods

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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 6

Quasi-Newton methods

- ▶ Mid 1950s, W. C. Davidon (Argonne National Laboratory): the first quasi-Newton algorithm, developed to accelerate a coordinate descent calculation.
- ▶ Fletcher and Powell: much faster and more reliable than the existing methods. Davidon's paper was not accepted for publication; it appeared in 1991.

The idea

Like steepest descent, use **only the gradient** at each iterate. Measure the **changes in gradients** and build from them a model of f that is good enough to give **superlinear convergence**.

No second derivatives are needed. This chapter: small and medium-sized problems.

The quadratic model and the iteration

Model of f at the current iterate x_k :

$$m_k(p) = f_k + \nabla f_k^T p + \frac{1}{2} p^T B_k p. \quad (6.1)$$

B_k is an $n \times n$ symmetric positive definite matrix, **updated** at every iteration. The minimizer of m_k is the search direction:

$$p_k = -B_k^{-1} \nabla f_k \quad (6.2), \quad x_{k+1} = x_k + \alpha_k p_k \quad (6.3),$$

where α_k satisfies the Wolfe conditions (3.6).

- ▶ This is the line search Newton method with B_k in place of the true Hessian.
- ▶ Direction from the model m_k ; step length from f itself (Wolfe).
- ▶ B_k is updated using the curvature measured during the most recent step.

The secant equation

New model at x_{k+1} : $m_{k+1}(p) = f_{k+1} + \nabla f_{k+1}^T p + \frac{1}{2} p^T B_{k+1} p$.

Requirement: ∇m_{k+1} matches ∇f at the latest two iterates x_k and x_{k+1} .

At x_{k+1} : $\nabla m_{k+1}(0) = \nabla f_{k+1}$, automatically. At x_k :

$$\nabla m_{k+1}(-\alpha_k p_k) = \nabla f_{k+1} - \alpha_k B_{k+1} p_k = \nabla f_k \implies B_{k+1} \alpha_k p_k = \nabla f_{k+1} - \nabla f_k. \quad (6.4)$$

Secant equation

$$s_k = x_{k+1} - x_k = \alpha_k p_k, \quad y_k = \nabla f_{k+1} - \nabla f_k, \quad (6.5)$$

$$B_{k+1} s_k = y_k. \quad (6.6)$$

The curvature condition

Premultiply (6.6) by s_k^T : a positive definite B_{k+1} can map s_k into y_k only if

$$s_k^T y_k > 0. \quad (6.7)$$

- ▶ f strongly convex: (6.7) holds for any two points x_k, x_{k+1} (Exercise 6.1).
- ▶ f nonconvex: (6.7) must be enforced by the line search.

The Wolfe conditions (3.6), or strong Wolfe (3.7), guarantee (6.7)

From (6.5) and (3.6b), $\nabla f_{k+1}^T s_k \geq c_2 \nabla f_k^T s_k$, and therefore

$$y_k^T s_k \geq (c_2 - 1) \alpha_k \nabla f_k^T p_k. \quad (6.8)$$

Since $c_2 < 1$ and p_k is a descent direction, the right-hand side is positive.

Determining B_{k+1} uniquely

When (6.7) holds, the secant equation (6.6) has infinitely many solutions: a symmetric matrix has $n(n+1)/2$ degrees of freedom, and the secant equation imposes n conditions.

Additional condition: among all symmetric matrices satisfying the secant equation, B_{k+1} is **closest to the current matrix B_k** :

$$\min_B \|B - B_k\| \quad (6.9a)$$

$$\text{subject to } B = B^T, \quad Bs_k = y_k. \quad (6.9b)$$

Each matrix norm in (6.9a) gives a different quasi-Newton method.

The weighted Frobenius norm

$$\|A\|_W \equiv \|W^{1/2}AW^{1/2}\|_F, \quad \|C\|_F^2 = \sum_{i=1}^n \sum_{j=1}^n c_{ij}^2. \quad (6.10)$$

W can be *any* matrix satisfying $Wy_k = s_k$. For concreteness, $W = \bar{G}_k^{-1}$, where $\bar{G}_k$ is the **average Hessian**

$$\bar{G}_k = \left[\int_0^1 \nabla^2 f(x_k + \tau \alpha_k p_k) d\tau \right]. \quad (6.11)$$

By Taylor's theorem (Theorem 2.1), $y_k = \bar{G}_k \alpha_k p_k = \bar{G}_k s_k$ (6.12).

With this W the norm is non-dimensional: the solution of (6.9) does not depend on the units of the problem.

Why a weighted norm? (Not in the book.)

$\|\cdot\|_W$ is an error metric that takes the curvature into account.

Example: $\nabla^2 f = \text{diag}(1000, 0.001)$ and $B = \nabla^2 f + I$.

- ▶ Direction e_1 : curvature 1001 instead of 1000. The step is almost exact.
- ▶ Direction e_2 : curvature 1.001 instead of 0.001. The step is 1000 times too short.

The Frobenius norm gives both errors the size 1. With $W = \nabla^2 f^{-1}$ they have sizes 10^{-3} and 10^3 , in proportion to their effect on the step.

Equivalently: in the variables $z = \bar{G}_k^{1/2} x$ the curvature is I in every direction, and $\|\cdot\|_W$ is the ordinary Frobenius norm there. The solution of (6.9) depends on W only through $Wy_k = s_k$, so $\bar{G}_k$ is never computed.

The DFP update

The unique solution of (6.9) in the norm (6.10), with $\rho_k = 1/(y_k^T s_k)$ (6.14):

$$\text{(DFP)} \quad B_{k+1} = (I - \rho_k y_k s_k^T) B_k (I - \rho_k s_k y_k^T) + \rho_k y_k y_k^T. \quad (6.13)$$

With $H_k = B_k^{-1}$ the direction (6.2) is a matrix–vector product. By the Sherman–Morrison–Woodbury formula (A.28):

$$\text{(DFP)} \quad H_{k+1} = H_k - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} + \frac{s_k s_k^T}{y_k^T s_k}. \quad (6.15)$$

- ▶ Davidon (1959); Fletcher and Powell. Both formulas are **rank-two** modifications: the matrix is not recomputed from scratch.
- ▶ The book states (6.13) without derivation. **Read:** pp. 138–139.

The BFGS update

Apply the same argument to H_k instead of B_k . The secant equation is now $H_{k+1}y_k = s_k$:

$$\min_H \|H - H_k\| \quad (6.16a)$$

$$\text{subject to } H = H^T, \quad Hy_k = s_k. \quad (6.16b)$$

Norm (6.10), now with $Ws_k = y_k$. The unique solution, with ρ_k from (6.14):

$$\text{(BFGS)} \quad H_{k+1} = (I - \rho_k s_k y_k^T) H_k (I - \rho_k y_k s_k^T) + \rho_k s_k s_k^T. \quad (6.17)$$

Broyden, Fletcher, Goldfarb, Shanno. Considered the most effective update.

BFGS in terms of B_k ; duality

Applying the Sherman–Morrison–Woodbury formula (A.28) to (6.17):

$$\text{(BFGS)} \quad B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{y_k^T s_k}. \quad (6.19)$$

Duality. The DFP and BFGS formulae are obtained from each other by the interchanges

$$s \leftrightarrow y, \quad B \leftrightarrow H.$$

	B_{k+1}	H_{k+1}
DFP	(6.13)	(6.15)
BFGS	(6.19)	(6.17)

Working with B_k : solving $B_k p_k = -\nabla f_k$ costs $O(n^3)$. Updating the factors $L_k D_k L_k^T$ of B_k costs $O(n^2)$, with no real advantage over Algorithm 6.1 in practice.

Algorithm 6.1 (BFGS method)

Given starting point x_0 , convergence tolerance $\epsilon > 0$,
inverse Hessian approximation H_0 ; $k \leftarrow 0$
while $\|\nabla f_k\| > \epsilon$
 compute the search direction $p_k = -H_k \nabla f_k$ (6.18)
 set $x_{k+1} = x_k + \alpha_k p_k$, where α_k satisfies the Wolfe conditions (3.6)
 define $s_k = x_{k+1} - x_k$ and $y_k = \nabla f_{k+1} - \nabla f_k$
 compute H_{k+1} by means of (6.17); $k \leftarrow k + 1$

- ▶ Cost per iteration: $O(n^2)$ arithmetic operations. No linear system is solved.
- ▶ Rate: superlinear. Newton's method is quadratic, but each iteration needs second derivatives and the solution of a linear system.

Numerical example: Rosenbrock's function (2.22)

Starting point $(-1.2, 1)$. Wolfe conditions on the step length in all three methods.

Iterations until $\|\nabla f_k\| \leq 10^{-5}$

steepest descent	5264
BFGS	34
Newton (inexact)	21

Last four values of $\|x_k - x^*\|$

steepest descent	BFGS	Newton
1.827e-04	1.70e-03	3.48e-02
1.826e-04	1.17e-03	1.44e-02
1.824e-04	1.34e-04	1.82e-04
1.823e-04	1.01e-06	1.17e-08

The superlinear rate of BFGS is visible in the last iterations. **Read: p. 141.**

Properties of the BFGS method

Positive definiteness is preserved

Let H_k be positive definite and $y_k^T s_k > 0$. For any nonzero z ,

$$z^T H_{k+1} z = w^T H_k w + \rho_k (z^T s_k)^2 \geq 0, \quad w = z - \rho_k y_k (s_k^T z).$$

Zero only if $s_k^T z = 0$; then $w = z \neq 0$ and the first term is positive. Hence H_{k+1} is positive definite.

Self-correction.

Read: pp. 141–142.

- ▶ If H_k estimates the curvature incorrectly and this slows down the iteration, H_k tends to correct itself within a few steps. DFP is less effective at this.
- ▶ The property holds only with an adequate line search (Wolfe).

BFGS versus DFP (Not in the book.)

Iterations; strong Wolfe line search ($c_1 = 10^{-4}$, $c_2 = 0.9$), $\alpha = 1$ tried first.

f	H_0	BFGS	DFP
quadratic, $n = 10$,	$10^{-4}/$	94	> 5000
eigenvalues 1 to 100,	$10^{-2}/$	45	249
$\ \nabla f_k\ < 10^{-8}$	$/$	11	15
Rosenbrock (2.22), $\ \nabla f_k\ < 10^{-5}$	$/$	34	84

- ▶ DFP changes B minimally, so B_k too large (steps too short) persists; a line search starting at $\alpha = 1$ does not repair this.
- ▶ BFGS changes H minimally: H_k too large (steps too long) is fixed by the line search.
- ▶ Theorem 6.5 extends to $\phi_k \in [0, 1)$ in (6.32), **but not to DFP** ($\phi_k = 1$).

Implementation: the line search

- ▶ Use the Wolfe conditions (3.6) or the strong Wolfe conditions (3.7), and **always try $\alpha_k = 1$ first**. This step length will eventually always be accepted (under certain conditions), which gives superlinear convergence.
- ▶ A fairly inaccurate line search is more economical in function evaluations. Commonly used: $c_1 = 10^{-4}$, $c_2 = 0.9$.
- ▶ Armijo backtracking from $\alpha = 1$ does not guarantee $y_k^T s_k > 0$: a step length greater than 1 may be required.
- ▶ Skipping the update ($H_{k+1} = H_k$) when $y_k^T s_k$ is negative or too close to zero is **not recommended**: the updates may be skipped much too often.

Read: pp. 142–143.

Implementation: the initial matrix H_0

$H_0 = \beta I$: there is no good general strategy for choosing β .

Heuristic: scale H_0 *after* the first step has been computed and *before* the first update:

$$H_0 \leftarrow \frac{y_k^T s_k}{y_k^T y_k} I. \quad (6.20)$$

The factor approximates an eigenvalue of $\nabla^2 f(x_k)^{-1}$, so (6.20) makes the size of H_0 similar to that of $\nabla^2 f(x_0)^{-1}$. **Read: (6.21), p. 143.**

(Not in the book.) With (6.20), the number of BFGS iterations hardly depends on the initial β : on the quadratic in the BFGS-versus-DFP table, 43–44 iterations for $\beta = 10^{-4}, 10^{-2}, 1$ (without (6.20): 94, 45, 11). (6.20) removes the choice of β ; it does not always reduce the iteration count.

The SR1 formulas

The only symmetric rank-one update that satisfies the secant equation:

$$(SR1) \quad B_{k+1} = B_k + \frac{(y_k - B_k s_k)(y_k - B_k s_k)^T}{(y_k - B_k s_k)^T s_k}. \quad (6.24)$$

By the Sherman–Morrison formula (A.27):

$$(SR1) \quad H_{k+1} = H_k + \frac{(s_k - H_k y_k)(s_k - H_k y_k)^T}{(s_k - H_k y_k)^T y_k}. \quad (6.25)$$

- ▶ Even if B_k is positive definite, B_{k+1} **may not be**. With trust-region methods, the ability to generate indefinite approximations is a chief advantage.
- ▶ Main drawback: **the denominator can vanish**, even for a convex quadratic.

Why SR1 is still of interest

- (i) A simple safeguard prevents breakdown and numerical instabilities.
- (ii) The matrices generated by the SR1 formula tend to be **good approximations to the true Hessian matrix**, often better than the BFGS approximations.
- (iii) In quasi-Newton methods for constrained problems, and for partially separable functions, it may not be possible to impose the curvature condition $y_k^T s_k > 0$, so BFGS updating is not recommended. In these settings indefinite Hessian approximations are desirable insofar as they reflect indefiniteness in the true Hessian.

B_k may be indefinite, so SR1 is used with a **trust region** (Algorithm 6.2), not a line search. **Read:** pp. 145–149 (Algorithm 6.2, Theorems 6.1, 6.2).

The skipping rule

Safeguard

Apply the update (6.24) only if

$$|s_k^T(y_k - B_k s_k)| \geq r \|s_k\| \|y_k - B_k s_k\|, \quad (6.26)$$

where $r \in (0, 1)$ is a small number, say $r = 10^{-8}$. Otherwise set $B_{k+1} = B_k$.

If $y_k \neq B_k s_k$ but $(y_k - B_k s_k)^T s_k = 0$, **no** SR1 update exists. Why is skipping acceptable for SR1, but not for BFGS?

- ▶ **SR1.** $s_k^T(y_k - B_k s_k) \approx 0$ occurs infrequently. It implies $s_k^T \bar{G} s_k \approx s_k^T B_k s_k$: the curvature of B_k along s_k is already correct.
- ▶ **BFGS.** Without Wolfe, $s_k^T y_k > 0$ fails easily; skipping can occur often.

The Broyden class

A family of updates with a scalar parameter ϕ_k :

$$B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{y_k^T s_k} + \phi_k (s_k^T B_k s_k) v_k v_k^T, \quad (6.32)$$

$$v_k = \left[\frac{y_k}{y_k^T s_k} - \frac{B_k s_k}{s_k^T B_k s_k} \right]. \quad (6.33)$$

$\phi_k = 0$: BFGS. $\phi_k = 1$: DFP. Equivalently, $B_{k+1} = (1 - \phi_k) B_{k+1}^{\text{BFGS}} + \phi_k B_{k+1}^{\text{DFP}}$.

All members satisfy the secant equation (6.6). **Restricted Broyden class:**

$\phi_k \in [0, 1]$; positive definiteness is preserved when $s_k^T y_k > 0$. **Read: Theorems 6.3, 6.4 (quadratic f), pp. 150–152.**

What is proved, and what is not

- ▶ BFGS and SR1 are remarkably robust in practice. For general nonlinear f , however, global convergence is not established: “it is not yet known if the algorithms enjoy such properties” (p. 153).
- ▶ (Not in the book.) For nonconvex f the BFGS iterates can fail to converge: examples of Dai (2002, Wolfe line search) and Mascarenhas (2004, exact line search).
- ▶ The analysis assumes that f is convex, or that the iterates satisfy certain properties. Tools: the trace and the determinant of B_k , combined in

$$\psi(B) = \text{trace}(B) - \ln(\det(B)). \quad (6.49)$$

Notation of §6.4: $\|\cdot\|$ is the Euclidean vector or matrix norm; $G(x) = \nabla^2 f(x)$.

Global convergence of BFGS: Theorem 6.5

Assumption 6.1

- (i) The objective function f is twice continuously differentiable.
- (ii) The level set $\mathcal{L} = \{x \in \mathbb{R}^n \mid f(x) \leq f(x_0)\}$ is convex, and there exist positive constants m and M such that $m\|z\|^2 \leq z^T G(x)z \leq M\|z\|^2$ (6.39) for all $z \in \mathbb{R}^n$ and $x \in \mathcal{L}$.

Theorem 6.5

Let B_0 be any symmetric positive definite initial matrix, and let x_0 be a starting point for which Assumption 6.1 is satisfied. Then the sequence $\{x_k\}$ generated by Algorithm 6.1 (with $\epsilon = 0$) converges to the minimizer x^* of f .

Read: proof of Theorem 6.5, pp. 154–156.

Superlinear convergence of BFGS: Theorem 6.6

Assumption 6.2

The Hessian matrix G is Lipschitz continuous at x^* , that is,

$$\|G(x) - G(x^*)\| \leq L\|x - x^*\|,$$

for all x near x^* , where L is a positive constant.

Theorem 6.6

Suppose that f is twice continuously differentiable and that the iterates generated by the BFGS algorithm converge to a minimizer x^* at which Assumption 6.2 holds. Suppose also that (6.52) holds. Then x_k converges to x^* at a superlinear rate.

Read: proof of Theorem 6.6, pp. 157–160.

Reading Theorem 6.6

- ▶ Applies to general nonlinear f , not just convex f ; $x_k \rightarrow x^*$ is an assumption.
- ▶ With $G_* = G(x^*)$, the proof establishes

$$\lim_{k \rightarrow \infty} \frac{\|(B_k - G_*)s_k\|}{\|s_k\|} = 0.$$

- ▶ This is the Dennis–Moré characterization (3.36). By Theorem 3.6, the unit step length $\alpha_k = 1$ satisfies the Wolfe conditions near the solution, and the rate of convergence is superlinear.
- ▶ $B_k \rightarrow \nabla^2 f(x^*)$ is not required. It suffices that B_k becomes increasingly accurate along the search directions (p. 47).

Read: SR1: Theorem 6.7, pp. 160–161.

Chapter 6 in one table

	BFGS	DFP	SR1
Update	rank two	rank two	rank one
Obtained from	(6.16), closest H	(6.9), closest B	(6.24)
Positive definite	if $s_k^T y_k > 0$	if $s_k^T y_k > 0$	not guaranteed
Breakdown	no	no	possible; skip by (6.26)
Self-correction	very effective	less effective	—
Framework	line search	line search	trust region
ϕ_k in (6.32)	0	1	may be outside $[0, 1]$
Global convergence	Thm. 6.5 (convex f)	not in Thm. 6.5	Thm. 6.7
Rate	superlinear (Thm. 6.6)	—	$(n+1)$ -step superlinear
$B_k \rightarrow \nabla^2 f(x^*)$	not required	—	Thm. 6.2

Three results to carry forward

1. **The secant equation** $B_{k+1}s_k = y_k$. Curvature information is obtained from differences of gradients. The update costs $O(n^2)$; no second derivatives and no linear system.
2. **The Wolfe conditions are part of the BFGS method.** They guarantee $s_k^T y_k > 0$, which keeps H_k positive definite and p_k a descent direction. Self-correction also depends on them.
3. **Superlinear convergence does not require** $B_k \rightarrow \nabla^2 f(x^*)$. Accuracy along the search directions is enough (BFGS, Theorem 6.6). SR1 does approximate the Hessian (Theorems 6.1, 6.2) and allows indefinite B_k within a trust region.

Proved in the book: Theorems 6.1, 6.5, 6.6.

Stated without proof: Theorems 6.2, 6.3, 6.4, 6.7.

Homework (problems 3–5 are not in the book)

1. Exercise 6.1(a): if f is strongly convex, then (6.7) holds for any x_k and x_{k+1} .
2. Exercise 6.2: the second strong Wolfe condition (3.7b) implies (6.7).
3. By hand. $f(x) = \frac{1}{2}x^T A x - b^T x$, $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $x_0 = 0$, $H_0 = I$. Run two iterations of Algorithm 6.1 with exact line searches. Check that $H_1 y_0 = s_0$, $x_2 = x^*$, $H_2 = A^{-1}$.
4. By hand. $B_k = I$, $s_k = (1, 0)^T$, $y_k = (-1, 0)^T$. Compute B_{k+1} from (6.24). Is it positive definite? Why can the BFGS update not be used here?
5. Implement Algorithm 6.1 with the scaling (6.20) and a library Wolfe line search. Minimize Rosenbrock's function (2.22) from $(-1.2, 1)$. Compare the last four values of $\|x_k - x^*\|$ with the table on p. 141.